HOMEWORK 2

ANTONY PEREZ

3.1. Multiply

a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 13 \\ 31 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \\ 2 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 11 \\ 4 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$ = impossible, matrix dimensions.

3.2. Let a linear transformation in $\mathbb{R}^2$ be a reflection in the line $x_1 = x_2$. Find the matrix.

Proof. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ □

3.3. For each linear transformation below, find its matrix:

a) $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x, y)^T = (x + 2y, 2x - 5y, 7y)^T$;
b) $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(x_1, x_2, x_3, x_4)^T = (x_1 + x_2 + x_3 + x_4, x_2 - x_4, x_1 + 3x_2 + 6x_4)^T$;
c) $T : \mathbb{P}_n \to \mathbb{P}_n, Tf(t) = f'(t)$;
d) $T : \mathbb{P}_n \to \mathbb{P}_n, Tf(t) = 2f(t) + 3f'(t) - 4f''(t)$.

Proof. **A.** $\begin{bmatrix} 1 & 2 \\ 2 & -5 \\ 0 & 7 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 3 & 0 & 4 \end{bmatrix}$ **C.** $\begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \ddots & n \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 & 3 & -8 & \dots & 0 \\ 0 & 2 & 6 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \ddots & 3n \\ 0 & 0 & \dots & \dots & 2 \end{bmatrix}$ □

3.4. Find 3×3 matrices representing the transformations of $\mathbb{R}^3$ which:

a) project every vector onto $x - y$ plane;
b) reflect every vector through $x - y$ plane;
c) rotate the $x - y$ plane through 30 deg, leaving z-axis alone.

Proof. **A.** $\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$ **B.** $\begin{bmatrix}1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$ **C.** $\begin{bmatrix}\cos 30&-\sin 30&0\\\sin 30&\cos 30&0\\0&0&1\end{bmatrix}$ □

3.5. Let A be a linear transformation. If z is the center of the straight interval $[x, y]$, show that Az is the center of the interval $[Ax, Ay]$.

Proof. Let $z = \frac{x+y}{2}$. Since A is a linear transformation, then z works by $\frac{1}{2}(Ax+Ay)$ through scalar multiplication and vector addition defined by the linear transformation. This would also represent the center of $[Ax, Ay]$. □

3.6. The set $\mathbb{C}$ of complex numbers can be identified with the space $\mathbb{R}^2$ by treating each $z = x + iy \in \mathbb{C}$ as a column $(x, y)^T \in \mathbb{R}^2$.

a) Treating $\mathbb{C}$ as a complex vector space, show that the multiplication by $\alpha = a + ib \in \mathbb{C}$ is a linear transformation in $\mathbb{C}$. What is its matrix?
b) Treating $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ show that the multiplication by $\alpha = a + ib$ defines a linear transformation there. What is its matrix?
c) Define $T(x + iy) = 2x - y + i(x - 3y)$. Show that this transformation is not a linear transformation in the complex vectors space $\mathbb{C}$, but if we treat $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ then it is a linear transformation there (i.e. that T is a *real linear* but not a *complex linear* transformation).
Find the matrix of the real linear transformation T.

Proof. **A.** It would be a 1x1 matrix of the form $\begin{bmatrix}1\end{bmatrix}$. $T(\alpha z) = \alpha T(z)$. $T(\alpha(u+z)) = \alpha T(u + z)$. **B.** The matrix is $\begin{bmatrix}1&0\\0&1\end{bmatrix}$. **C.** This isn't possible due to the 1x1 matrix limitation set by $\mathbb{C}$. And, since A cannot contain any variables, this is immediately false under the supposition that T over $\mathbb{C}$ is a linear transformation. And that T is a linear transformation under $\mathbb{R}^2$, given by $A = \begin{bmatrix}2&-1\\1&-3\end{bmatrix}$ □

3.7. Show that any linear transformation in $\mathbb{C}$ is a multiplication by $\alpha \in \mathbb{C}$.

Proof. The basis of $\mathbb{C}$ is the 1x1 standard basis. Thus, we let $A = [1]$. Since $\alpha \in \mathbb{C}$, we can represent α as $x + iy$. Now, we may show that $T(1) = \alpha$, making this statement true, since now we can show that $T(z\alpha) = z\alpha$. □

5.1. Let

$$A = \begin{pmatrix}1&2\\3&1\end{pmatrix}, \quad B = \begin{pmatrix}1&0&2\\3&1&-2\end{pmatrix}, \quad C = \begin{pmatrix}1&-2&3\\-2&1&-1\end{pmatrix}, \quad D = \begin{pmatrix}-2\\2\\1\end{pmatrix}$$

a) Mark all the products that are defined, and give the dimensions of the result: $AB, BA, ABC, ABD, BC, BC^T, B^TC, DC, D^TC^T$.
b) Compute $AB, A(3B + C), B^TA, A(BD), (AB)D$.

Proof. $A \in M_{2\times 2}$, $B, C \in M_{2\times 3}$, $D \in M_{3\times 1}$.
$AB = 2 \times 3$
BA = undefined
ABC = undefined
$ABD = 2 \times 1$
BC = undefined
$BC^T = 2 \times 2$

$B^TC = 3 \times 3$
$DC =$ undefined
$D^TC^T = 1 \times 2$

$$AB = \begin{pmatrix} 7 & 2 & -2 \\ 6 & 1 & 4 \end{pmatrix}.$$

$$A(3B + C) = \begin{pmatrix} 18 & 6 & -5 \\ 19 & -2 & 20 \end{pmatrix}.$$

$$B^TA = \begin{pmatrix} 10 & 5 \\ 3 & 1 \\ -4 & 2 \end{pmatrix}.$$

$$A(BD) = \begin{pmatrix} -8 \\ -4 \end{pmatrix}.$$

$$(AB)D = \begin{pmatrix} -8 \\ -4 \end{pmatrix}.$$

□

5.2. Let T_γ be the matrix of rotation by γ in $\mathbb{R}^2$. Check by matrix multiplication that $T_\gamma T_{-\gamma} = T_{-\gamma}T_\gamma = I$

Proof. Let $T_\gamma = \begin{pmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{pmatrix}$ and $T_{-\gamma} = \begin{pmatrix} \cos\gamma & \sin\gamma \\ -\sin\gamma & \cos\gamma \end{pmatrix}$. Either matrix can be multiplied commutatively because of the properties of cosine and sine, giving us the identity $\cos^2(\gamma) + \sin^2(\gamma) = 1$. □

5.3. Multiply two rotation matrices T_α and T_β (it is a rare case when the multiplication is commutative, i.e. $T_\alpha T_\beta = T_\beta T_\alpha$, so the order is not essential). Deduce formulas for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$ from here.

Proof. $\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix} = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & \cos\alpha\cos\beta - \sin\alpha\sin\beta \end{pmatrix}$
You're essentially rotating by β, then rotating by α, therefore $T_{\alpha+\beta}$ is the same thing, which would give us the geometric formulas asked of us. □

5.4. Find the matrix of the orthogonal projection in $\mathbb{R}^2$ onto the line $x_1 = -2x_2$.

Proof. $\frac{1}{5}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$ □

5.5. Find linear transformations $A, B : \mathbb{R}^2 \to \mathbb{R}^2$ such that $AB = \mathbf{0}$ but $BA \neq \mathbf{0}$.

Proof. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ □

5.6. Prove that $\text{trace}(AB) = \text{trace}(BA)$.

Proof. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} aw + by & ax + bz \\ cw + dy & xc + dz \end{bmatrix}$. $\text{Trace}(AB) = ax + by + xc + dz$
$\begin{bmatrix} w & x \\ y & z \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} wa + xc & wb + xd \\ ya + zc & yb + zd \end{bmatrix}$ $\text{Trace}(BA) = wa + xc + yb + zd.$
Therefore, both traces are equivalent. □

5.7. Construct a non-zero matrix A such that $A^2 = 0$.

Proof. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $A^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{bmatrix}$. We know that a, b, c, d cannot all be simultaneously zero, but A^2 must be zero. We can deduce that a, d are zero, and either b, c can be zero, but not both, or else this would violate our non-zero matrix. Let $b = 0$. Now, we see $A^2 = 0$, and c is the only entry that isn't a zero in A. □

5.8. Find the matrix of the reflection through the line $y = -2x/3$. Perform all the multiplications.

Proof. $\frac{1}{13}\begin{bmatrix} 3 & 2 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 3 & -2 \\ 2 & -3 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} 5 & -12 \\ -12 & -5 \end{bmatrix}$ □